

# SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-26 22:22 UTC

## SEC P vs NP — notebook summary

Generated 2026-04-26 22:22 UTC

- Cycles recorded: **119**
- Time span: 72.7h (~1.64 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp\_notebook.jsonl

### Verdict distribution

| Verdict      | Count |
|--------------|-------|
| INCONCLUSIVE | 100   |
| FALSIFIED    | 14    |
| SUPPORTED    | 4     |
| BARRIER_HIT  | 1     |

### Mathematical fields explored (field\_A)

| Field                                                                             | Cycles |
|-----------------------------------------------------------------------------------|--------|
| Ergodic Circuit Framework (communication complexity via dynamical systems)        | 4      |
| {“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”} | 4      |
| Fourier analysis of boolean functions                                             | 3      |
| Noncommutative geometry                                                           | 2      |
| Representation theory of symmetric groups                                         | 2      |
| Ergodic Circuit Framework (COMM_COMPLEXITY)                                       | 2      |
| Matroid Theory                                                                    | 2      |
| Tropical geometry                                                                 | 2      |
| Combinatorial homotopy theory                                                     | 1      |
| Kazhdan-Lusztig theory of Hecke algebras                                          | 1      |
| Combinatorial algebraic topology                                                  | 1      |
| Knot theory (quantum invariants)                                                  | 1      |
| Quadratic forms over finite fields (Grothendieck-Witt groups)                     | 1      |
| Clifford algebras over finite fields                                              | 1      |
| Toric geometry (via Gröbner degenerations)                                        | 1      |
| Arithmetic geometry (Tate-Shafarevich groups)                                     | 1      |
| Geometric Langlands correspondence                                                | 1      |

| Field                                                              | Cycles |
|--------------------------------------------------------------------|--------|
| Algebraic graph theory (chromatic polynomials)                     | 1      |
| Clifford algebras over real vector spaces                          | 1      |
| Algebraic cycles (Chow groups over finite fields)                  | 1      |
| Algebraic lattice theory (Möbius functions of flow lattices)       | 1      |
| Directed algebraic topology (d-space homology)                     | 1      |
| Motivic integration (Denef-Loeser zeta functions)                  | 1      |
| Matroid theory (Clifford index of binary matroids)                 | 1      |
| Representation theory of finite groups (Frobenius-Schur indicator) | 1      |
| Categorification and knot Floer homology                           | 1      |
| Modular forms                                                      | 1      |
| Algebraic geometry (ideals in polynomial rings)                    | 1      |
| Hypergraph Tutte polynomials                                       | 1      |
| Galois theory of finite field extensions                           | 1      |

### Last 15 cycles (chronological)

| Time                 | Verdict      | Title                                                                                       |
|----------------------|--------------|---------------------------------------------------------------------------------------------|
| 2026-04-26 17:10 UTC | INCONCLUSIVE | Nisan-Wigderson Seed Length Bounded by Finite Geometry Line Count                           |
| 2026-04-26 17:44 UTC | INCONCLUSIVE | Persistent Homology Bottleneck Distance                                                     |
| 2026-04-26 18:14 UTC | INCONCLUSIVE | Lower-Bounds Set-Disjoint Antichain Cover Number of KW Relation Lower-Bounds Monotone Formu |
| 2026-04-26 18:40 UTC | INCONCLUSIVE | Tropical Lipschitz Stability of Discrepancy under Fourier Coeffic                           |
| 2026-04-26 19:10 UTC | INCONCLUSIVE | Young-Tableau Cell Count Bounds Border-Rank of 3x3 Permanent Slic                           |
| 2026-04-26 19:40 UTC | INCONCLUSIVE | Lehmer Pair Density of Communication Matrix Lower-Bounds Discrepa                           |
| 2026-04-26 20:32 UTC | FALSIFIED    | Tropical Self-Convolution Doubling Law for MinimalFourierCoeffici                           |
| 2026-04-26 20:50 UTC | FALSIFIED    | Tropical Shift-Invariance of MinimalFourierCoefficient under Addi                           |
| 2026-04-26 20:54 UTC | INCONCLUSIVE | Hypercontractive (4,2)-Norm of Clause-Count Polynomial Bounds Res                           |
| 2026-04-26 21:31 UTC | INCONCLUSIVE | Tropical Tensor-Product Factorization of MinimalFourierCoefficien                           |

| Time                 | Verdict      | Title                                                                   |
|----------------------|--------------|-------------------------------------------------------------------------|
| 2026-04-26 21:34 UTC | INCONCLUSIVE | Dyadic Martingale Quadratic<br>Variation Lower-Bounds<br>Sign-Rank Disc |
| 2026-04-26 21:43 UTC | INCONCLUSIVE | Legendre-Fenchel Involution<br>Conjecture: Discrepancy<br>Invariance un |
| 2026-04-26 21:46 UTC | INCONCLUSIVE | Combinatorial Discrepancy of<br>NW Designs Predicts<br>Generator Bias o |
| 2026-04-26 22:20 UTC | FALSIFIED    | Tropical Max-Aggregation<br>Monotonicity of<br>MinimalFourierCoefficien |
| 2026-04-26 22:22 UTC | BARRIER_HIT  | Schur-Horn Majorization Gap<br>Lower-Bounds Sign-Rank<br>Discrepancy    |

## How to read the reports

- reports/supported\_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified\_findings.md — FALSIFIED with counterexamples.
- reports/daily\_YYYY-MM-DD.md — chronological log by day.
- pvsnp\_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

## PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```